

Preregistered hypotheses

Preregistered hypotheses I

The registration in `data/example_registration.json` declares its hypotheses, outcomes, exclusion rules, and analysis plan before any result is observed. It is frozen under a SHA-256 content hash (prefix 96a34a11d132); the validator in `registered_report.protocol.validate_registration` confirms the payload carries every required section with no structural findings.

Confirmatory hypothesis I

- ▶ **H1** — *The proposed condition improves the primary score.*
This is the sole confirmatory hypothesis. It maps to the registered outcome `primary_score`, measured as a mean score difference and analysed with a two-sided label-permutation test at `alpha = 0.05`.

Each registered hypothesis is bound to exactly one registered outcome and one registered analysis. `[@fig:hypothesis_map]` renders that binding directly from the frozen registration via `hypothesis_outcome_map`, so the figure cannot silently drift from the locked plan.

Confirmatory hypothesis II

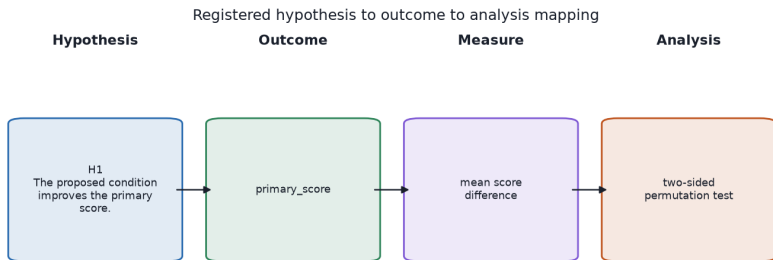


Figure 1: Registered hypothesis-to-outcome-to-analysis mapping for the demonstration registration. Each row is generated by `hypothesis_outcome_map(registration)` in `src/registered_report/demo_study.py`: hypothesis H1 links to the `primary_score` outcome, the mean score difference measure, and the two-sided permutation test analysis. Only registered hypotheses appear; exploratory endpoints are excluded by construction.

Registered outcome and exclusion rule I

- ▶ **Primary outcome** `primary_score` — mean score difference between the proposed and reference conditions, evaluated by a two-sided permutation test.
- ▶ **Exclusion rule** — records missing the primary outcome are excluded *before* group assignment is inspected, so exclusion decisions cannot depend on the observed effect.

No secondary outcome is preregistered. Any additional endpoint introduced during analysis is therefore exploratory and must be recorded in the deviation ledger ([Section @sec:deviations]) rather than presented as a confirmatory finding.